Research Test Memorandum for DeepSeek

Sharp Hypercontractivity for a Two-Parameter Nontracial Qubit Semigroup

Problem Brief and Staged Research Instructions

6 November 2026

Abstract

This memorandum is the problem brief to be supplied directly to DeepSeek. It states the mathematical setting, the target questions, the required standards of proof, and a staged research plan. The principal objective is to determine the sharp weighted $L_2(\tau) \to L_q(\tau)$ contraction region of a two-parameter phase-covariant qubit channel and to deduce the optimal hypercontractive time of the associated semigroup. The recommended strategy begins with fixed numerical and symbolic parameter regimes, then progresses to general q, general bias μ, tensor powers, and finally selected $p \neq 2$ extensions. The model must clearly separate theorems, conditional arguments, numerical evidence, and conjectures.

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1 Mission and expected deliverable

You are asked to conduct a time-limited mathematical investigation of a sharp noncommutative hypercontractivity problem. The final submission should contain:

1. a precise statement of every result proved;
2. complete proofs, with all reductions and boundary cases justified;
3. counterexamples if a proposed statement is false;
4. reproducible numerical experiments used to guide or falsify conjectures;
5. a clear list of unresolved steps;
6. a literature-dependence statement distinguishing known ingredients from new arguments.

Do not describe a feasible estimate as optimal unless a matching necessary condition or extremizing sequence is supplied. Do not treat a local Taylor expansion or numerical optimization as a global proof.

Primary target. Determine the exact two-parameter region in which the qubit map defined below is a contraction from weighted $L_2(\tau)$ to weighted $L_q(\tau)$, and use that region to compute the optimal hypercontractive time of the specified semigroup.

2 Precise mathematical setting

2.1 State and weighted noncommutative L_p spaces

Fix a bias parameter

$$0 < \mu < \frac{1}{2}$$

and the faithful density matrix

$$\tau = \tau_\mu := \begin{pmatrix} \mu & 0 \\ 0 & 1-\mu \end{pmatrix}. \tag{1}$$

For $1 \le p < \infty$, define

$$\|X\|_{p,\tau} := \left(\operatorname{Tr}\left|a u^{1/(2p)} X \tau^{1/(2p)}\right|^p\right)^{1/p}, \qquad X \in M_2(\mathbb{C}). \tag{2}$$

Use the operator norm when $p = \infty$. All logarithms are natural. Every proof must use the convention (2); if another representation of the weighted space is introduced, prove the equivalence being used.

2.2 Two-parameter channel

For real parameters ρ, θ, define

$$\Phi_{\rho,\theta}\begin{pmatrix} x & z \\ w & y \end{pmatrix} = \begin{pmatrix} m + \rho(x-m) & \theta z \\ \theta w & m + \rho(y-m) \end{pmatrix}, \qquad m = \mu x + (1-\mu) y. \tag{3}$$

The map fixes the identity, preserves the τ-expectation, scales the centered diagonal direction by ρ, and scales the off-diagonal coherence directions by θ.

Before using channel terminology, determine the exact completely positive parameter domain of (3). Keep the following sets distinct:

1. the domain in which $\Phi_{\rho,\theta}$ is completely positive and unital;
2. the domain in which it is merely a linear $L_2(\tau) \to L_q(\tau)$ contraction;
3. the one-dimensional parameter path generated by the semigroup below.

2.3 Continuous-time semigroup

Fix

$$\frac{1}{2} < k < 1, \tag{4}$$

and define

$$T_t := \Phi_{e^{-t},e^{-kt}}, \qquad t \geq 0. \tag{5}$$

Equivalently,

$$T_t \begin{pmatrix} x & z \\ w & y \end{pmatrix} = \begin{pmatrix} m + e^{-t}(x-m) & e^{-kt}z \\ e^{-kt}w & m + e^{-t}(y-m) \end{pmatrix}.$$

The generator has eigenvalues 0, 1, and k on the constant, centered diagonal, and coherence sectors. The $L_2(\tau)$ spectral gap is k. A general logarithmic-Sobolev comparison therefore supplies a non-sharp benchmark, but the objective is the exact curve rather than a comparison estimate.

3 Main quantities and questions

Definition 3.1 (Single-site contraction region)**.** *For $q \geq 2$, define*

$$\mathcal{R}^{(1)}_{2,q}(\mu) := \left\{ (\rho,\theta) : \|\Phi_{\rho,\theta}(X)\|_{q,\tau} \leq \|X\|_{2,\tau} \text{ for every } X \in M_2(\mathbb{C}) \right\}. \tag{6}$$

Definition 3.2 (Optimal hypercontractive time)**.**

$$t_*(2,q;\mu,k) := \inf \left\{ t \geq 0 : \|T_t\|_{L_2(\tau)\to L_q(\tau)} \leq 1 \right\}. \tag{7}$$

3.1 Diagonal threshold

The restriction of (3) to the diagonal algebra is the biased two-point noise operator. Its sharp $2 \to q$ threshold is

$$\rho_{\mu,q} = \left[\frac{(1-\mu)^{2/q} - \mu^{2/q}}{\mu^{2/q-1}(1-\mu) - (1-\mu)^{2/q-1}\mu} \right]^{1/2}. \tag{8}$$

Thus $\rho \leq \rho_{\mu,q}$ is a necessary condition for membership in (6). You may use the classical biased two-point result only if stated accurately and matched to the normalization above.

3.2 Candidate coherence threshold

A candidate coherence threshold suggested by the sharp qubit depolarizing argument is

$$\theta_{\mu,q} = \left[\frac{[4\mu(1-\mu)]^{1/2-1/q}}{q-1} \right]^{1/2}. \tag{9}$$

At $\mu = 1/2$, this becomes $(q-1)^{-1/2}$. For nontracial τ, the identity perturbation $I + \varepsilon A$ generally detects only the simpler spectral-gap threshold and does not by itself establish (9).

Candidate conjecture 3.3 (Rectangular-region conjecture)**.** *Within the completely positive parameter domain,*

$$\mathcal{R}_{2,q}^{(1)}(\mu) \stackrel{?}{=} [0, \rho_{\mu,q}] \times [0, \theta_{\mu,q}]. \tag{10}$$

If (10) is valid, then

$$t_*(2, q; \mu, k) = \max\left\{-\log \rho_{\mu,q}, -\frac{1}{k}\log \theta_{\mu,q}\right\}. \tag{11}$$

The main issue is whether mixed diagonal-coherence inputs create an additional coupled boundary beyond the two separate thresholds.

4 Required research tasks

4.1 Task A: establish the parameter foundations

Problem 4.1 (Complete positivity and reversibility)**.** *Determine the exact completely positive parameter domain for $\Phi_{\rho,\theta}$. Verify unitality, τ-preservation, and the relevant symmetry or reversibility property. Prove that the path (e^{-t}, e^{-kt}) lies in that domain for the stated range of k.*

Problem 4.2 (Necessary conditions)**.** *Derive the diagonal condition $\rho \le \rho_{\mu,q}$. Construct a coherence-sensitive family and determine the strongest necessary condition it yields for θ. Decide whether the exact necessary condition is (9).*

4.2 Task B: solve the single-site optimization

Problem 4.3 (Rectangular region or coupled boundary)**.** *Prove or disprove Conjecture 3.3. The argument must cover every complex 2×2 matrix. A restriction to positive, self-adjoint, or normal matrices must be justified by a precise reduction.*

A satisfactory solution must address:

1. whether the norm can be reduced to positive matrices by polar decomposition, 2-positivity, or another valid principle;
2. how the output q-norm changes when the magnitude of an off-diagonal entry is increased;
3. whether the sign of the relevant derivative is uniform in the diagonal entries and in ρ;
4. rank-deficient boundary points and noninteger q;
5. the possibility of a mixed extremizer with both diagonal imbalance and nonzero coherence.

Problem 4.4 (Sharpness and equality)**.** *Classify diagonal extremizers, coherent extremizing families, limiting extremizers near the identity, and any mixed equality cases. Distinguish an attained maximum from a sharp constant obtained only by an extremizing sequence.*

4.3 Task C: deduce the semigroup curve

After determining the single-site region, compute the first intersection of

$$t \longmapsto (e^{-t}, e^{-kt})$$

with that region. Prove the resulting formula for $t_*(2, q; \mu, k)$ and analyze which obstruction dominates as (μ, k, q) vary. Identify any transition surface defined by equality of the diagonal and coherence times.

4.4 Task D: tensor stability

For $n \geq 1$, define

$$\mathcal{R}_{2,q}^{(n)}(\mu) := \left\{ (\rho, \theta) : \left\| \Phi_{\rho,\theta}^{\otimes n}(X) \right\|_{q,\tau^{\otimes n}} \leq \|X\|_{2,\tau^{\otimes n}} \text{ for every } X \in M_2^{\otimes n} \right\}. \tag{12}$$

Investigate

$$\mathcal{R}_{2,q}^{(n)}(\mu) \stackrel{?}{=} \mathcal{R}_{2,q}^{(1)}(\mu). \tag{13}$$

Begin with $n = 2$ and $q = 3$ or $q = 4$. Product-input tests are not sufficient: the argument or counterexample search must include entangled operators. If a PODS or norm-compression theorem is invoked, verify line by line that independent parameters ρ and θ are permitted.

4.5 Task E: optional $p \neq 2$ extension

For a fixed pair such as

$$(p, q) = \left(\frac{4}{3}, 4 \right) \quad \text{or} \quad (p, q) = \left(\frac{3}{2}, 3 \right),$$

define $\mathcal{R}_{p,q}^{(1)}(\mu)$ by replacing the input norm in (6). Determine whether the region remains rectangular or develops a coupled boundary

$$\theta \leq \Theta_{\mu,p,q}(\rho).$$

This task should be attempted only after the $p = 2$ structure is understood.

5 Recommended staged research plan

The following progression is recommended. Do not wait for a complete general proof before reporting a rigorous subcase.

5.1 Stage 0: sanity checks and exact algebra

First verify:

1. the complete-positivity domain from the Choi matrix;
2. the $L_2(\tau)$ orthogonal decomposition into constants, centered diagonal matrices, and coherences;
3. the tracial limit $\mu = 1/2$;
4. the trivial endpoint $q = 2$;
5. the semigroup and generator identities.

These calculations should be completed before attempting any general norm inequality.

5.2 Stage 1: fixed parameters

Begin with concrete values that allow direct symbolic and numerical checking:

$$\mu = \frac{1}{4}, \qquad q = 4, \qquad k = \frac{3}{4}.$$

Recommended questions:

1. Is the proposed single-site region correct at $q = 4$?
2. Can the 4-norm be expanded into a polynomial expression that makes the mixed terms explicit?
3. Does numerical optimization find a violation near a corner of the proposed rectangle?
4. Are all observed maximizers diagonal, coherent, or infinitesimal perturbations?

A complete proof for this fixed case is a valid intermediate theorem. If $q = 4$ is still difficult, first restrict to positive matrices and then prove or disprove the reduction from general matrices.

5.3 Stage 2: one-parameter symbolic families

After the fixed case, vary one parameter at a time:

1. arbitrary μ with $q = 4$;
2. arbitrary even integer q with a fixed rational μ;
3. arbitrary $q \geq 2$ in the tracial case $\mu = 1/2$;
4. the standard depolarizing line $\rho = \theta$ as a consistency check;
5. the pure diagonal edge $\theta = 0$ and the maximal coherence edge allowed by complete positivity.

The purpose is to identify which part of the proof genuinely depends on q, which part depends on the bias, and which part depends on the equality or independence of the two damping parameters.

5.4 Stage 3: general single-site theorem

Attempt the full result for arbitrary

$$0 < \mu < \frac{1}{2}, \qquad q \geq 2.$$

A promising route is to parameterize a positive input as

$$X = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}, \qquad a, c \geq 0, \qquad |b|^2 \leq ac,$$

and study the output norm as a function of $s = |b|^2$. Derive the exact Fréchet or eigenvalue formula needed for noninteger q. Determine whether the derivative in s is controlled uniformly by (9). If positivity is insufficient, return to a singular-value or block-matrix reduction for general X.

5.5 Stage 4: semigroup phase diagram

Assuming the region is known, compare

$$t_{\mathrm{diag}}(\mu, q) = -\log \rho_{\mu,q}$$

and

$$t_{\mathrm{coh}}(\mu, k, q) = -\frac{1}{k} \log \theta_{\mu,q}.$$

Determine where each dominates. Plot or characterize the transition equation

$$t_{\mathrm{diag}}(\mu, q) = t_{\mathrm{coh}}(\mu, k, q).$$

Analyze the limits $\mu \downarrow 0$, $\mu \uparrow 1/2$, $q \downarrow 2$, and $q \to \infty$.

5.6 Stage 5: tensor powers

Proceed in the order:

1. $n = 2$, $q = 4$;
2. arbitrary n, $q = 4$;
3. integer $q \geq 3$;
4. real $q > 2$ via interpolation, only after verifying that interpolation preserves the proposed sharp boundary.

At each stage, test entangled inputs numerically. Identify whether the tensor proof uses only coordinatewise bounds or whether a coupling between ρ and θ appears.

5.7 Stage 6: general $p \to q$

Only after completing or substantially understanding $2 \to q$, study fixed $p \neq 2$. Expect new mixed terms because the input is no longer governed by a Hilbert norm. The correct outcome may be a nonrectangular region. A rigorous counterexample to rectangularity is as valuable as a positive theorem.

6 Numerical investigation requirements

Numerics are a falsification and discovery tool, not a proof. Use homogeneity to impose $\|X\|_{2,\tau} = 1$. Search over:

1. positive matrices;
2. self-adjoint indefinite matrices;
3. general complex matrices through singular-value decomposition or eight real coordinates;
4. boundary and nearly rank-one inputs;
5. small perturbations of the identity;
6. genuinely mixed inputs with unequal diagonal entries and nonzero coherence.

Use multistart global optimization, record random seeds, and recompute candidate violations at high precision. Recommended initial grids are

$$\mu \in \{0.05, 0.15, 0.25, 0.40, 0.49\}, \quad q \in \{2.25, 3, 4, 6, 10\},$$

and parameter points just inside and just outside each proposed boundary. If a violation is found, provide the explicit matrix, parameter values, objective ratio, precision, and a rigorous or interval-certified follow-up if possible.

7 Permitted conclusions and reporting discipline

Use the following labels exactly:

- **Theorem:** every logical step is proved;
- **Conditional theorem:** the result follows from one explicitly isolated unproved lemma;
- **Computer-assisted theorem:** numerical enclosures and error bounds are rigorous and reproducible;
- **Numerical evidence:** high-precision computation without proof;
- **Conjecture:** a proposed statement supported by reasoning or experiments;
- **Counterexample:** an explicit object whose defining inequalities are verified.

A final answer must not conceal a gap by phrases such as "standard", "clearly", or "by convexity" unless the exact theorem and its hypotheses are supplied.

8 Minimum acceptable and stretch outcomes

8.1 Minimum rigorous outcome

A useful minimum submission contains:

1. the complete-positivity domain;
2. the two necessary thresholds;
3. a complete proof for one fixed case such as $(\mu, q) = (1/4, 4)$, or a rigorous counterexample to the rectangular conjecture;
4. reproducible full-matrix numerical tests;
5. a precise identification of the remaining lemma needed for the general case.

8.2 Strong outcome

A strong submission proves the single-site rectangular region for every $0 < \mu < 1/2$ and $q \geq 2$, derives (11), and classifies extremizers.

8.3 Exceptional outcome

An exceptional submission additionally proves lossless tensorization, or finds and rigorously characterizes a tensor-level obstruction. A substantive fixed $p \neq 2$ theorem or counterexample is also exceptional.

9 Literature use

The following results are relevant background:

- finite-dimensional quantum logarithmic Sobolev inequalities and hypercontractivity [1];
- hypercontractivity for self-adjoint unital qubit semigroups in the tracial setting [2];
- PODS norm compression, tensorization, and sharp generalized qubit depolarizing estimates [3];
- the geometry of phase-covariant qubit channels [4].

The standard generalized depolarizing channel has a single damping parameter. Do not assume that its sharp theorem automatically applies to independent ρ and θ. If an existing theorem is claimed to settle any task in this memorandum, quote its exact statement and verify every hypothesis.

10 Final checklist

Before submitting, confirm that the report answers all applicable questions below:

1. What is the exact completely positive parameter domain?
2. What is rigorously known about $\mathcal{R}^{(1)}_{2,q}(\mu)$?
3. Is the rectangular conjecture proved, disproved, or unresolved?
4. Are all reductions from general matrices justified?
5. Are boundary and rank-deficient cases covered?
6. What are the extremizers or extremizing sequences?
7. What is the resulting formula or bound for $t_*(2, q; \mu, k)$?
8. What happens for $n = 2$ and for general tensor powers?
9. Which statements are inherited from the literature, and which are new?
10. Can every numerical claim be reproduced from the supplied code and seeds?

References

[1] M. J. Kastoryano and K. Temme, Quantum logarithmic Sobolev inequalities and rapid mixing, *Journal of Mathematical Physics* **54** (2013), 052202. https://doi.org/10.1063/1.4804995.

[2] C. King, Hypercontractivity for Semigroups of Unital Qubit Channels, *Communications in Mathematical Physics* **328** (2014), 285–301. https://doi.org/10.1007/s00220-014-1982-4.

[3] Y. Dong, L. Gao, F. Ou, P. Yao, and H. Zhou, Dimension-Free Approximate Tensorization of Quantum Hypercontractivity for Qudit Depolarizing Semigroups, *arXiv:2606.17729*, version 2, 2026. https://arxiv.org/abs/2606.17729.

[4] K. Siudzińska, Geometry of phase-covariant qubit channels, *Journal of Physics Communications* **7** (2023), 075002. https://doi.org/10.1088/2399-6528/ace0f4.